



FLEET & SHORE PERSONNEL TRAINING

Safe Welding Training

Welding hazards onboard, the hot-work permit system, PPE and equipment safety, and how to prepare a work area and respond if something goes wrong.



Hot Work Permit

SOLAS Ch. II-2

Company SMS

ISM Code

What We Will Cover

01

Welding Hazards Onboard

Fire, fumes, arc flash and electric shock

02

The Hot-Work Permit System

How permits are issued, controlled and closed out

03

PPE for Welding

Protecting eyes, skin and airways from arc and fume

04

Preparing a Safe Work Area

Gas-freeing, fire watch and enclosed-space precautions

05

Welding-Equipment Safety

Gas cylinders, regulators, hoses and electrical leads

06

Emergency Response to Welding Incidents

Fire, burns, electric shock and fume exposure



SECTION 1

Welding Hazards Onboard

Why welding is treated as one of the highest-risk routine jobs carried out on a ship.

01

WHY IT MATTERS

A Concentration of Serious Hazards

Welding combines several hazards that are individually dangerous and, together, have caused some of the most severe shipboard casualties.

- Open flame and intense heat from arc or gas welding can ignite fuel, oil residue, paint, insulation or gas vapour in the surrounding area.
- Welding fume contains fine particulates and metal oxides that are harmful to the lungs with repeated or prolonged exposure.
- Arc flash produces intense ultraviolet and infrared radiation that can burn unprotected skin and cause permanent eye damage within seconds.
- Electric shock risk is present throughout arc welding, particularly in damp, confined or metallic spaces where the body forms part of the circuit.
- A large proportion of serious shipboard fires begin as hot-work incidents — most are preventable with correct planning and precautions.



Never Assume It's Safe

A space that looks empty and non-flammable can still hold vapour, residue or an oxygen-deficient atmosphere — every hot-work job starts from a proper risk assessment, not assumption.

THE CORE HAZARDS

Four Hazards Every Welder Must Manage

Each hazard requires its own specific controls, but all must be addressed together on every job.



Fire & Explosion

Sparks, hot slag and radiant heat can ignite flammable material or vapour well beyond the immediate work area.



Fume & Gas Exposure

Welding fume, shielding gas displacement and burning coatings (paint, galvanising) can create a toxic or oxygen-deficient atmosphere.



Arc Flash & Radiation

UV and IR radiation from the arc causes skin burns and 'arc eye' (photokeratitis) without adequate shielding.



Electric Shock

Welding sets carry high current; damaged leads, wet conditions or poor earthing sharply increase shock risk.



SECTION 2

The Hot-Work Permit System

02

No welding, cutting or grinding takes place aboard without a valid, signed hot-work permit.

WHY A PERMIT SYSTEM

Controlling Hot Work Through a Permit

A hot-work permit forces a documented risk assessment and sign-off before any spark-producing work begins.

- The permit records the exact location, the work to be done, the hazards identified, and the precautions required before work starts.
- It is issued by a responsible officer (typically the Chief Officer or Chief Engineer) after inspecting the work area and confirming controls are in place.
- The permit is valid only for the stated location, task and time window — any change in scope or location requires a new or revised permit.
- A gas-free certificate, where the space requires one, must be attached to or referenced by the permit before it is issued.
- The permit is displayed at the worksite and only closed out once the job is complete, the area re-inspected and the fire watch stood down.



No Permit, No Work

This is an absolute rule with no informal exceptions — even a five-minute grinding job requires a valid permit if it produces sparks or heat.

WHO DOES WHAT

Roles in the Permit Process

Clear responsibility at each stage keeps the permit system effective rather than a paperwork formality.

Role	Responsibility
Permit Issuer (senior officer)	Assesses the risk, sets precautions, issues and ultimately closes the permit
Welder / Person doing the work	Confirms understanding of the permit conditions and stops work if conditions change
Fire Watch	Remains at the worksite throughout hot work and for a set period after, watching for smouldering material
Master / Duty Officer	Retains overall authority to suspend hot work at any time if conditions become unsafe

Using the Permit Correctly

The permit is only as good as the discipline applied to it on the day.



Do

- Read and understand every precaution listed before starting work
- Stop immediately and reassess if conditions change (e.g. new cargo, wind shift, gas readings rise)
- Keep the permit and gas-free certificate visible at the worksite
- Complete the close-out inspection and fire watch period before leaving the area unattended



Don't

- Don't begin any spark- or heat-producing work without a signed permit
- Don't extend the permit's scope, location or time without a new authorisation
- Don't leave the fire watch post unattended during or immediately after hot work
- Don't treat the permit as a formality once work is under way



SECTION 3

PPE for Welding

03

The protective equipment that shields the welder from arc radiation, heat, spatter and fume.

Protecting Eyes, Skin and Airways

Welding PPE is task-specific — the correct combination depends on the process, position and environment.



Welding Helmet / Shield

Fitted with the correct filter shade for the process and current, protecting eyes and face from arc radiation and spatter.



Flame-Resistant Clothing

Leather or treated flame-resistant overalls, gauntlets and apron protect against spatter, sparks and radiant heat.



Respiratory Protection

Fume extraction at source, supplemented by a respirator in confined or poorly ventilated spaces.



Foot & Hand Protection

Insulated welding gloves and non-conductive, spark-resistant safety boots without exposed metal fastenings.

CORRECT USE

Wearing and Maintaining Welding PPE

PPE only protects if it is in good condition and worn correctly for the whole job.



Do

- Inspect helmet lens, gloves and clothing for damage before every job
- Select the correct filter shade for the welding process and current being used
- Keep clothing free of oil, grease or flammable residue that could ignite from spatter
- Remove or cover flammable items (lighters, matches) before starting work



Don't

- Don't weld with a cracked lens, damaged gloves or torn clothing
- Don't wear synthetic clothing that can melt onto the skin if ignited
- Don't roll up sleeves or leave skin exposed near the arc or hot metal
- Don't rely on PPE alone — it supplements, not replaces, other controls



SECTION 4

Preparing a Safe Work Area

Gas-freeing, fire watch arrangements and enclosed-space precautions that must be in place before the arc is struck.

04

AREA PREPARATION

Clearing and Securing the Worksite

The work area must be prepared and verified safe before the permit is issued, not after work has started.

- All flammable material — rags, paint, oil, gas cylinders not in use — is removed from the area or shielded with fire-resistant covers.
- Adjacent compartments, especially those on the other side of the bulkhead or deck being worked, are checked for heat transfer risk.
- Fixed firefighting systems in the space are confirmed operational, and portable extinguishers of the correct type are positioned at the worksite.
- Openings, drains and cable runs that could carry sparks or heat to another space are sealed or covered before work begins.
- Ventilation is arranged to clear welding fume without introducing a flammable atmosphere from an adjacent space.



Test the Atmosphere

A calibrated gas meter must confirm the atmosphere before entry and hot work in any enclosed or previously sealed space — never rely on smell or visual inspection alone.

GAS-FREE STANDARD

What 'Gas-Free' Actually Means

A gas-free certificate confirms specific, measured conditions — not simply that the space 'seems fine.'

20.9%

Oxygen content required — neither oxygen-deficient nor oxygen-enriched

0%

Flammable vapour reading on the combustible gas indicator

TLV

Toxic gas readings kept below the relevant Threshold Limit Value throughout the work

Hot Work in Enclosed or Confined Spaces

Enclosed-space entry procedures apply in full whenever hot work is carried out inside a tank, void or similar space.



Do

- Complete the enclosed-space entry checklist and permit alongside the hot-work permit
- Maintain continuous mechanical ventilation throughout the work, not only before entry
- Keep a dedicated attendant at the entrance with rescue equipment ready
- Re-test the atmosphere periodically throughout an extended job



Don't

- Don't enter or weld in an enclosed space on a single, one-time gas reading
- Don't rely on natural ventilation alone for an enclosed space
- Don't allow the attendant to leave the entrance while anyone remains inside
- Don't assume a space is safe because it was gas-free on a previous occasion



SECTION 5

Welding-Equipment Safety

Gas cylinders, regulators, hoses and electrical leads — the equipment that must be checked before every job.

05

Cylinders, Regulators and Hoses

Oxy-acetylene equipment carries its own specific hazards separate from the arc itself.

- Gas cylinders are always secured upright, in a well-ventilated stowage, with oxygen and acetylene kept physically separated.
- Cylinder valves, regulators and hoses are checked for damage, leaks and correct fitting before each job — a leak test uses soapy water, never a flame.
- Flashback arrestors are fitted at both the regulator and torch end of oxy-acetylene hoses to prevent a flame travelling back into the cylinder.
- Hoses are colour-coded (typically red for fuel gas, blue or black for oxygen) and never interchanged between services.
- Empty and full cylinders are stored separately and clearly marked to prevent an empty cylinder being connected in error.



Acetylene Instability

Acetylene becomes unstable above about 1 bar gauge pressure — regulators and equipment rated for acetylene must never be substituted with equipment rated only for other gases.

Welding Sets and Electrical Leads

Electrical faults in welding equipment are a leading cause of both shock injuries and equipment-started fires.

- Welding leads and cables are inspected for cut, cracked or worn insulation before connecting the set — damaged leads are taken out of service immediately.
- The earth/return clamp is connected as close as practical to the work to minimise stray current through the ship's structure.
- Welding sets are properly earthed and, where required, protected by a voltage-reduction device to lower open-circuit voltage when not welding.
- Cables are run clear of walkways, sharp edges and other hot-work sparks, and are never coiled tightly around metal during operation.
- Equipment is isolated and de-energised before any maintenance, and connections are only made or broken with the power source switched off.

PRE-JOB CHECKS

Equipment Checks Before Every Job

A short, consistent equipment check catches most equipment-related incidents before they happen.



Cylinders & Gauges

Secured, valves and gauges undamaged, correct regulator fitted, no oil or grease on oxygen fittings.



Hoses & Connections

No cuts, cracks or leaks; flashback arrestors fitted; connections tight and correctly matched.



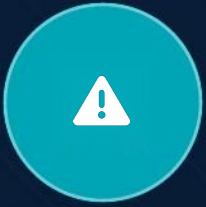
Electrical Leads

Insulation intact, connections secure, earth clamp properly attached close to the work.



Torch / Electrode Holder

Free of damage, correctly assembled, and matched to the gas type or electrode being used.



SECTION 6

Emergency Response to Welding Incidents

What every crew member does in the first seconds of a welding fire, burn, shock or fume-exposure incident.

06

If a Fire Starts

The fire watch and nearby crew are the first line of response to any fire that develops from hot work.



Do

- Raise the alarm immediately, even for a small or apparently controlled fire
- Cut off the gas or electrical supply to the welding equipment if safe to do so
- Use the correct type of extinguisher already positioned at the worksite
- Continue the fire watch for the full required period after the fire is out, checking for reignition



Don't

- Don't delay raising the alarm while attempting to fight the fire alone
- Don't use water on an electrical or fuel-fed fire
- Don't re-enter a space to retrieve equipment once evacuation has begun
- Don't stand down the fire watch early — smoldering material can reignite well after work stops

Burns, Shock and Fume Exposure

Each type of welding injury requires a specific first response before further medical treatment.

- For a suspected electric shock, isolate the power source before touching the casualty — never touch them while they remain in contact with a live circuit.
- Cool a thermal burn with clean running water as soon as possible and cover loosely with a sterile, non-adhesive dressing; do not apply ice directly.
- For arc eye (photokeratitis), remove the casualty from further UV exposure, do not rub the eyes, and seek medical attention even if symptoms seem mild.
- For suspected fume or gas overexposure, move the casualty to fresh air immediately and administer oxygen and first aid per the ship's medical guide.
- Report every welding-related injury or near miss, however minor, so the Company SMS can identify and correct any recurring cause.



Call for Help Early

For any burn beyond superficial, suspected fume overexposure, or an electric shock casualty, contact the ship's medical officer and consider Telemedical Assistance without delay.



COURSE COMPLETE

A valid permit, the right PPE and a properly prepared space turn a high-risk job into a controlled one.

Company DPA — 24/7 emergency line

Fleet Technical Department — mng@gloryship.com.tr

Fleet Safety Department — mng@gloryship.com.tr